SERII DE NUMERE REALE. PROBLEME PROPUSE

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Să se determine natura şi suma seriilor următoare :

1. $\sum_{n=1}^{\infty} \frac{1}{4n^2-1}$
2. $\sum_{n=1}^{\infty} \frac{\sqrt{2n+1}-\sqrt{2n-1}}{\sqrt{4n^2-1}}$
3. $\sum_{n=0}^{\infty} \operatorname{arctg} \frac{1}{n^2+3n+3}$
4. $\sum_{n=1}^{\infty} (\sqrt{n+\alpha+1}-2\sqrt{n+\alpha}+\sqrt{n+\alpha-1}), \alpha > 0$
5. $\sum_{n=1}^{\infty} \frac{1}{(\alpha+n)(\alpha+n+1)}$, α fiind un număr real diferit de orice întreg negativ.
6. $\sum_{n=1}^{\infty} \frac{1}{16n^2-8n-3}$
7. $\sum_{n=1}^{\infty} \ln \frac{n+1}{n}$

Să se stabilească natura seriilor următoare :

8. $\sum_{n=1}^{\infty} \arcsin \frac{n+1}{2n+3}$
9. $\sum_{n=1}^{\infty} \frac{2^n+3^n}{2^{n+1}+3^{n+1}}$
10. $\sum_{n=1}^{\infty} n^2 \cdot \ln\left(1+\frac{1}{n^2}\right)$
11. $\sum_{n=2}^{\infty} \frac{\ln(n+1)}{n^3}$
12. $\sum_{n=1}^{\infty} \frac{1}{7^n+3^n}$
13. $\sum_{n=0}^{\infty} 3^n \sin \frac{\pi}{5^n}$

14. $\sum_{n=1}^{\infty} \frac{n+3\sqrt{n}}{n(n+1)\sqrt{n+2}}$

15. $\sum_{n=1}^{\infty} \frac{1}{a^n+n}, \ \ a>0$

16. $\sum_{n=1}^{\infty} \frac{\ln n}{2n^3-1}$

17. $\sum_{n=1}^{\infty} \frac{1}{n^\alpha}(\sqrt[3]{n^2+n+1}-\sqrt[3]{n^2-n+1}), \ \ \alpha \in \mathbb{R}$

18. $\sum_{n=1}^{\infty} \frac{\sin \frac{1}{n(n+1)}}{\cos \frac{1}{n} \cos \frac{1}{n+1}}$

19. $\sum_{n=1}^{\infty} n^2(\mathrm{e}^{\frac{1}{n}}-1-\frac{1}{n})$

20. $\sum_{n=1}^{\infty} \mathrm{e}^{-n^2}$

21. $\sum_{n=1}^{\infty} n\mathrm{e}^{-(n^2+n)}$

22. $\sum_{n=1}^{\infty} \frac{(an)^n}{n!}$

23. $\sum_{n=0}^{\infty} \frac{3n^3-n^2+1}{n!}a^n, a>0$

24. $\sum_{n=1}^{\infty} a^n \left(1+\frac{1}{n}\right)^n, a>0$

25. $\sum_{n=1}^{\infty} \left(b \cdot \frac{a+n}{a+n-1}\right)^n, b>0$

26. $\sum_{n=1}^{\infty} \left(a \cdot \frac{n^2+n+1}{n^2}\right)^n, a>0$

27. $\sum_{n=1}^{\infty} \frac{a^n}{\sqrt{n!}}, a>0$

28. $\sum_{n=1}^{\infty} \left[\frac{(2n-1)!!}{(2n)!!}\right]^2 \cdot \frac{1}{n}$

29. $\sum_{n=1}^{\infty} \left(\frac{1 \cdot 4 \cdot 7 \cdot \ldots (3n-2)}{3 \cdot 6 \cdot 9 \ldots (3n)}\right)^2$

30. $\sum_{n=2}^{\infty} \frac{1}{(\ln \ln n)^2}$

31. $\sum_{n=1}^{\infty} \frac{\ln n}{\sqrt[3]{n^4}}$

32. $\sum_{n=1}^{\infty} \frac{1}{n^{\sqrt{n}}}$

33. $\sum_{n=2}^{\infty} \frac{1}{n \ln^2 n}$

34. $\sum_{n=1}^{\infty} \frac{1}{n \sqrt[n]{n}}$

35. $\sum_{n=1}^{\infty} \left[\frac{1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \cdot \ldots \cdot (2n)}\right]^{\alpha}, \alpha \in \mathbb{R}$

36. $\sum_{n=1}^{\infty} a^{\ln n}, a > 0$

37. $\sum_{n=1}^{\infty} \frac{\sqrt{n!}}{(2+\sqrt{1})(2+\sqrt{2}) \ldots (2+\sqrt{n})}$

38. $\sum_{n=2}^{\infty} \left(\frac{1}{\ln n}\right)^{\ln(\ln n)}$

39. $\sum_{n=1}^{\infty} [\sqrt{(n+1)(n+a)} - n]^n, a > 0$

40. $\sum_{n=1}^{\infty} \left(\frac{an+b}{cn+d}\right)^n, a, b, c, d > 0$

41. $\sum_{n=1}^{\infty} \mathrm{tg}^{\, n}(a + \frac{b}{n^2}), 0 < a < \frac{\pi}{2}, b \in \mathbb{R}$

42. $\sum_{n=2}^{\infty}(n-1)^2 \arcsin \frac{\pi}{2^{n+1}}$

43. $\sum_{n=1}^{\infty}2^{n-1} \cdot \operatorname{tg}^2 \frac{a}{2^n} \cdot \operatorname{tg} \frac{a}{2^{n-1}}, a > 0$

44. $\sum_{n=1}^{\infty}\frac{1}{4^n \cdot \cos^2 \frac{a}{2^n}}$

45. $\sum_{n=1}^{\infty}\frac{n^p \ln n}{n!}, p \in \mathbb{R}$

46. $\sum_{n=1}^{\infty}\sin \frac{\pi}{\sqrt{n}} \cdot \operatorname{tg} \frac{\pi}{\sqrt{n+1}}$

47. $\sum_{n=1}^{\infty}\sin \frac{1}{\pi n} \cdot \arcsin \frac{1}{\sqrt{2n^2+1}}$

48. $\sum_{n=1}^{\infty}\frac{a^n \ln n}{n}, a > 0$

49. $\sum_{n=1}^{\infty}(2n+1)\left[\frac{a(a-1)\ldots(a-n+1)}{(a+1)(a+2)\ldots(a+n)}\right]^2, a \in \mathbb{R} \setminus \mathbb{Z}$

50. $\sum_{n=1}^{\infty}\frac{n^p}{a+n3^n}, a > 0, p \in \mathbb{R}$

51. $\sum_{n=1}^{\infty}\sqrt{\frac{a^{n+1}+b}{(2a)^n}}, a, b > 1$

52. $\sum_{n=1}^{\infty}\frac{(n!)^2}{2^{n^2}}$

53. $\sum_{n=1}^{\infty}\left(\frac{1}{2^n} \cdot \operatorname{tg} \frac{a}{2^n}\right)^2$

54. $\sum_{n=1}^{\infty}a^{1+\frac{1}{2}+\frac{1}{3}+\ldots+\frac{1}{n}}, a > 0$

55. $\sum_{n=2}^{\infty}(-1)^n \frac{\sqrt{n+1}}{(n^2+n+1)^\alpha} \cdot \frac{1}{\ln^\beta(n+2)}, \alpha, \beta \in \mathbb{R}_+$

56. $\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n} + 1}{n}$

57. Să se arate că suma dintre o serie convergentă şi una divergentă este o serie divergentă .

58. Să se studieze convergenţa absolută şi convergenţa seriilor:
a) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{\sqrt{n(n+1)}}$; b) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{2n+1}{3^n}$; c) $\sum_{n=1}^{\infty} (-1)^n \frac{2^n + \text{sh}n}{3^n}$;
d) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{(n+1)^{n+1}}{n^{n+2}}$; e) $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{27n^2 + 36n + 11}{(3n+1)(3n+2)(3n+3)}$;
f) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} \sqrt[3]{n}}{n + \sqrt{7}}$